

Cross-Level Residual Frequency-Guided Multi-Scale Refinement with a ConvNeXt Encoder for Polyp Segmentation

Abstract

Accurate colorectal polyp segmentation remains challenging because lesions vary in scale, contrast, and boundary appearance. We present a ConvNeXt-Tiny U-shaped framework with bottleneck Frequency-Aware Feature Enhancement (FAFEM) and residual frequency-guided coupling to a Multi-Scale Convolution Block (MSCB). The resulting RFG-MSCB uses FAFEM’s low- and high-frequency descriptor to weight multi-kernel branches at the Stage-3 encoder-decoder fusion while preserving ordinary MSCB aggregation as a reference. Under a PraNet-style protocol, checkpoint selection uses only an internal validation subset, with Kvasir-SEG and CVC-ClinicDB subsets reserved for final seen-domain testing and three datasets used for external testing. In the five-dataset, three-seed ablation, the sequence from Baseline through FAFEM and FAFEM + MSCB to Residual RFG-MSCB culminated in the highest means for all six positive-valued metrics, including mDice 0.872 and mE_ϕ 0.941, while the displayed MAE remained 0.018 after adding FAFEM.

1 Introduction

Colorectal polyps are abnormal tissue growths that can develop into malignant lesions if they are not detected and removed at an early stage. Colonoscopy is widely used for the examination of the lower gastrointestinal tract and enables clinicians to identify and remove suspicious lesions. Nevertheless, visual inspection remains challenging because polyp appearance varies considerably across patients and imaging conditions. Variations in lesion size, morphology, texture, contrast, and location, together with specular reflections, motion blur, intestinal folds, bubbles, and other visual artifacts, can complicate accurate delineation of polyp regions. Automated polyp segmentation therefore remains an important problem in computer-aided colonoscopy, where reliable pixel-level localization may assist subsequent analysis and clinical decision support.

Accurate polyp segmentation requires a model to simultaneously capture high-level semantic information and preserve sufficiently detailed spatial representations. U-shaped encoder-decoder architectures are well suited to this task because they combine progressively abstract encoder features with higher-resolution representations through skip connections. U-Net established this general design, while subsequent architectures such as UNet++ further demonstrated the importance of multi-level feature interaction and hierarchical decoding [1, 2]. However, repeated downsampling reduces spatial resolution, and fixed decoder operations may not always provide sufficient flexibility for lesions that vary considerably in scale and visual frequency content.

Recent transformer-based and hybrid architectures have improved long-range dependency modeling in medical image segmentation. TransUNet combines convolutional representations with transformer encoding, while Polyp-PVT applies hierarchical transformer features and cross-level aggregation to polyp segmentation [3, 4]. These approaches demonstrate the importance of broad contextual modeling, but they can introduce additional architectural complexity. ConvNeXt provides a convolutional alternative with hierarchical representations and large-kernel depthwise spatial processing [5].

Multi-scale representation is particularly important in polyp segmentation because lesions may occupy very small regions or extend over a large portion of the image. Multi-Scale Convolution Blocks (MSCBs) address this issue by combining convolutional branches with different receptive fields, allowing a network to aggregate

local and broader spatial responses [6]. In their ordinary form, the branch responses are summed without sample-specific frequency conditioning.

Frequency-aware feature processing provides another complementary perspective. We adopt a locally implemented bottleneck FAFEM that uses local low-pass and residual high-pass proxies to enhance deep features and produce a compact descriptor. The descriptor can then provide image-dependent information for guiding later multi-scale processing rather than being used only at the bottleneck.

This observation motivates the central idea of this work. Instead of treating bottleneck frequency enhancement and decoder multi-scale refinement as independent operations, we establish an explicit cross-level interaction between them. The frequency descriptor produced by a bottleneck FAFEM is propagated to the first decoder stage and used to regulate the relative contributions of the multi-scale branches in an MSCB. In this way, the receptive-field mixture of the decoder becomes sample dependent rather than fixed across all images.

Directly replacing the original MSCB aggregation with a fully frequency-guided formulation could, however, unnecessarily perturb a stable multi-scale representation. To address this issue, we introduce the bounded residual guidance strategy formalized in Eq. (23). Here, M_{ord} denotes the ordinary multi-scale aggregation, M_{guide} denotes the frequency-conditioned aggregation, and s is a bounded learnable guidance-strength parameter. This formulation preserves the conventional MSCB pathway as a reference and allows the model to progressively learn frequency-dependent deviations during optimization.

Based on this mechanism, we develop a ConvNeXt-Tiny U-shaped segmentation framework in which the frequency-guided modification is localized to the first encoder–decoder fusion stage; the remaining decoder stages retain conventional fusion and depthwise-separable convolution.

The principal contributions of this work are summarized as follows:

- We introduce RFG-MSCB, which uses the bottleneck FAFEM descriptor to modulate the multi-kernel MSCB branches through bounded residual interpolation around ordinary MSCB aggregation.
- We evaluate the resulting ConvNeXt-Tiny segmentation architecture on five public colorectal polyp datasets using a comparison of Baseline, FAFEM, FAFEM + MSCB, and Residual RFG-MSCB configurations.
- We quantify the architectural trade-off across the four-configuration ablation: the localized RFG-MSCB modification adds about 0.074 M parameters over FAFEM + MSCB, with effectively unchanged GFLOPs at the displayed precision and modest matched-hardware runtime overhead.

2 Related Work

2.1 Encoder–decoder architectures for polyp segmentation

Fully convolutional encoder–decoder networks remain a central design pattern for biomedical segmentation. U-Net combines semantic context with spatial detail through skip connections, while UNet++ uses nested dense pathways to reduce the semantic gap between encoder and decoder representations [1, 2]. Polyp-specific convolutional networks subsequently developed this pattern in different directions: HarDNet-MSEG couples a low-memory-traffic backbone with lightweight decoding, whereas PraNet combines a parallel partial decoder with reverse attention for progressive region refinement [7, 8]. More recent designs emphasize cross-attention and feature exploration, multi-level lesion awareness, progressive coarse-to-fine decoding, or dual-view feature propagation [9, 10, 11, 12]. These studies demonstrate the value of cross-level reuse and staged refinement for heterogeneous polyp appearance. In contrast, our formulation uses a deep frequency descriptor to control the receptive-field mixture at an earlier decoder fusion.

Modern convolutional encoders offer a complementary way to strengthen this design without replacing the full hierarchy by self-attention. ConvNeXt modernizes a ConvNet using large-kernel depthwise convolution, inverted channel expansion, Layer Normalization, GELU, layer scaling, and stochastic depth [5]. Its

hierarchical outputs are suitable for U-shaped feature reuse, while depthwise spatial operators also support economical dense prediction. We use ConvNeXt-Tiny as an established encoder and focus on coupling bottleneck frequency processing with Stage-3 decoder refinement.

2.2 Transformer and hybrid contextual segmentation

Transformer and hybrid models address the broad variation in polyp size, morphology, and contrast by combining local appearance with long-range context. TransUNet provides an influential CNN–transformer formulation for medical segmentation, and Polyp-PVT adapts hierarchical pyramid-transformer features to polyp segmentation [3, 4]. Recent polyp methods extend this direction through contrastive transformer representations in CTNet, lightweight vision transformers in Polyp-LVT, contextual information flow in CIF-Former, and phase-wise feature-pyramid retention in PFPRNet [13, 14, 15, 16]. Multi-feature pyramid fusion has likewise been combined with visual transformers in MF-Net [17]. Collectively, these methods prioritize global dependency modeling and cross-scale semantic exchange, but often employ attention or transformer machinery throughout substantial parts of the network. Our method instead retains a convolutional ConvNeXt encoder–decoder and introduces one lightweight, descriptor- conditioned multi-scale refinement at Stage 3.

2.3 Efficient multi-scale convolutional decoding

Multi-scale decoding is particularly relevant to polyps, which may be small and low-contrast or occupy much of the endoscopic frame. Parallel kernels provide different effective receptive fields, while depthwise convolution limits the cost of spatial filtering. EMCAD formalized this principle in an efficient medical-image decoder whose Multi-Scale Convolution Blocks (MSCBs) expand the feature, process it with multiple depthwise kernels, aggregate the branch responses, shuffle channels, and project the result [6]. We adopt the lightweight MSCB topology as the ordinary multi-scale reference path rather than presenting it as a new block.

The broader adaptive-convolution literature shows that receptive fields need not be combined uniformly. Selective Kernel Networks use a feature descriptor to select among kernel scales, and Dynamic Convolution forms input-dependent mixtures over convolution kernels [18, 19]. Unlike these same-level routing mechanisms, RFG-MSCB transfers a low/high-frequency descriptor from the bottleneck to the Stage-3 decoder fusion. This cross-level signal weights the adopted MSCB’s 1×1 , 3×3 , and 5×5 depthwise branches while retaining their ordinary sum as the reference response. Recent medical segmentation designs that combine frequency enhancement with multi-scale adaptive attention further support the utility of jointly modeling spectral content and scale selection [20].

2.4 Frequency-aware and adaptive feature recalibration

Compact descriptors are widely used to recalibrate convolutional representations. Squeeze-and-Excitation Networks learn channel-wise gates from global pooled features, whereas FcaNet interprets channel attention in terms of selected frequency components [21, 22]. Frequency-aware processing has also moved beyond channel attention: Frequency-Aware Feature Fusion addresses the loss of high-frequency detail and inconsistency introduced when fusing features of different resolutions in dense prediction [23]. Recent medical segmentation methods have also combined frequency representations with cross-feature interaction and shape-oriented frequency partitioning [24, 25]. In polyp segmentation, Polyp-Mamba combines multi-frequency perception with gated selection, providing a domain-specific example of frequency-sensitive contextual modeling [26].

The present design differs in the source, destination, and form of its conditioning. The adopted bottleneck FAFEM produces local low-pass and residual high-pass responses and a pooled descriptor $\mathbf{z}_f$, which RFG-MSCB propagates across levels to control the Stage-3 multi-kernel mixture rather than recalibrating only the source feature. Building on frequency-aware processing and adaptive multi-scale aggregation, RFG-MSCB transfers the bottleneck descriptor across levels and uses the branch-conditioned response as a bounded residual refinement of conventional MSCB.

2.5 Boundary-aware and recent polyp-specific refinement

Weak boundaries and ambiguous foreground appearance have motivated complementary polyp-specific refinement strategies. MEGANet applies multi-scale edge-guided attention, while MSBP-Net predicts boundaries at multiple scales [27, 28]. MLB-Net combines lesion-aware and boundary-enhanced processing, and PraNet-V2 extends reverse attention with dual supervision [10, 29]. Reliability has also received increasing attention: MEIN models multi-scale uncertainty through evidence integration rather than treating all predictions as equally certain [30]. Together with CFFormer, PFPRNet, MSPN, and DVFIP-Net, these 2025–2026 methods illustrate a shift toward explicit contextual flow, hierarchical refinement, uncertainty modeling, and targeted boundary recovery [15, 16, 11, 12].

Boundary emphasis in our architecture is supplied by the training objective rather than a dedicated edge branch, contour decoder, or uncertainty head. The architectural focus is instead on conditioning receptive-field selection at a specific decoder level using bottleneck frequency composition while preserving an efficient multi-scale path. Accordingly, the method adopts ConvNeXt-Tiny, a locally implemented FAFEM, and the MSCB structure from EMCAD, then links the latter two through the cross-level bounded residual frequency-guided mechanism developed next.

3 Method

3.1 Overall Architecture

The network uses a U-shaped encoder–decoder architecture built upon a ConvNeXt-Tiny encoder. Frequency information extracted from deep semantic features regulates multi-scale processing during decoding. Frequency enhancement and multi-scale convolution are linked through a cross-level connection between the bottleneck and the first decoder stage.

Let $I \in \mathbb{R}^{3 \times H \times W}$ denote an input colonoscopy image. The ConvNeXt-Tiny encoder extracts four hierarchical feature maps,

$$\{F_1, F_2, F_3, F_4\} = E(I), \quad (1)$$

where $F_1 \in \mathbb{R}^{96 \times H/4 \times W/4}$, $F_2 \in \mathbb{R}^{192 \times H/8 \times W/8}$, $F_3 \in \mathbb{R}^{384 \times H/16 \times W/16}$, and $F_4 \in \mathbb{R}^{768 \times H/32 \times W/32}$. For an input resolution of 352×352 , their spatial dimensions are 88×88 , 44×44 , 22×22 , and 11×11 , respectively.

The deepest encoder feature F_4 is first processed by a Frequency-Aware Feature Enhancement Module (FAFEM), which produces an enhanced bottleneck representation together with a compact frequency descriptor. The enhanced representation is passed through a lightweight residual bottleneck, whereas the frequency descriptor is propagated to the first decoder stage. After the decoder feature is fused with F_3 , a Residual Frequency-Guided Multi-Scale Convolution Block (RFG-MSCB) uses the frequency descriptor to adaptively regulate its multi-scale convolution branches.

The overall processing can therefore be summarized as

$$(\tilde{F}_4, \mathbf{z}_f) = \mathcal{F}_{\text{FAFEM}}(F_4), \quad (2)$$

$$B = \mathcal{B}(\tilde{F}_4), \quad (3)$$

$$D_4 = \mathcal{M}_{\text{RFG}}(\mathcal{S}_3[\mathcal{D}_4(B), F_3], \mathbf{z}_f). \quad (4)$$

followed by

$$D_3 = \mathcal{S}_2 [\mathcal{D}_3(D_4), F_2], \quad (5)$$

$$D_2 = \mathcal{S}_1 [\mathcal{D}_2(D_3), F_1], \quad (6)$$

$$D_1 = \mathcal{D}_1(D_2). \quad (7)$$

Here, $\mathcal{D}_i$ denotes a decoder block, $\mathcal{S}_i$ denotes the corresponding skip-feature fusion operation, $\mathbf{z}_f$ is the frequency descriptor extracted at the bottleneck, and $\mathcal{M}_{\text{RFG}}$ denotes the residual frequency-guided multi-scale refinement mechanism. Equations (4)–(7) summarize the decoder path.

Figure 2 summarizes the complete data flow. The input image passes through the hierarchical ConvNeXt-Tiny encoder to produce F_1 – F_4 . At the bottleneck, FAFEM transforms the deepest feature F_4 into an enhanced representation and the 1536-dimensional descriptor $\mathbf{z}_f$; the enhanced representation then passes through the lightweight residual bottleneck and the fourth decoder block. After fusion with the Stage-3 encoder feature F_3 , RFG-MSCB uses $\mathbf{z}_f$ to modulate its multi-kernel branches. Thus, the descriptor is transferred from the bottleneck to a later decoder fusion rather than being used only locally. The ordinary MSCB aggregation remains the reference path, and the frequency-conditioned response enters as a bounded residual deviation. The remaining decoder stages progressively recover spatial resolution before the prediction head produces a single-channel logit map, which is bilinearly upsampled to form the final segmentation mask.

3.2 ConvNeXt-Tiny Encoder

The encoder employs ConvNeXt-Tiny with stage depths [3, 3, 9, 3] and channel dimensions [96, 192, 384, 768]. Before entering the encoder, each RGB image is normalized once using the ImageNet statistics $\mu = (0.485, 0.456, 0.406)$ and $\sigma = (0.229, 0.224, 0.225)$.

Figure 1 summarizes the stage-level organization and the internal structure of an individual ConvNeXt block. The first stage begins with a 4×4 stride-4 stem convolution, whereas Stages 2–4 begin with a 2×2 stride-2 downsampling convolution. Each stage then applies its prescribed number of blocks. Within each block, a 7×7 stride-1 depthwise convolution is followed by channels-last Layer Normalization and a pointwise MLP that expands C to $4C$, applies GELU, and projects $4C$ back to C . Layer scaling and stochastic depth are applied before the transformed feature is added to the residual path.

For an input feature X , the block transformation is formalized as

$$\begin{aligned} Z &= \text{DWConv}_{7 \times 7}(X), \\ V &= W_2 (\text{GELU} (W_1 (\text{LN}(Z)))) . \end{aligned} \quad (8)$$

The block output is obtained through layer-scaled residual learning,

$$Y = X + \text{DropPath}(\gamma \odot V), \quad (9)$$

where γ is a learnable layer-scale parameter initialized to 10^{-6} .

A stochastic-depth rate of 0.25 is used throughout the encoder. Feature dropout with probability 0.2 is additionally applied after the second, third, and fourth encoder stages.

3.3 Frequency-Aware Bottleneck Enhancement

The deepest encoder representation F_4 contains strong semantic information but has undergone substantial spatial downsampling. To expose complementary frequency characteristics of this representation, FAFEM is applied exclusively at the bottleneck.

Given $X = F_4$, a local low-frequency approximation is obtained using 3×3 average pooling with stride 1 and padding 1,

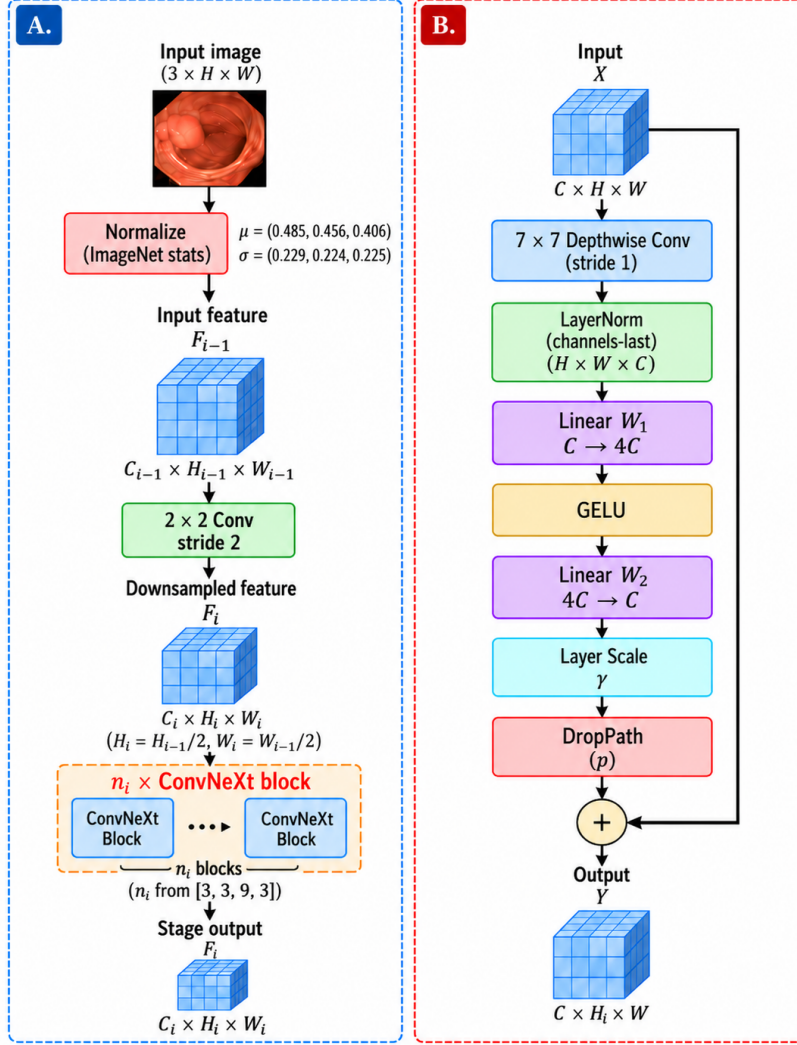


Figure 1: ConvNeXt-Tiny encoder structure. (A) Stage-level processing after the one-time ImageNet normalization: Stage 1 uses a 4×4 stride-4 stem, while subsequent stages use 2×2 stride-2 downsampling and repeat ConvNeXt blocks according to depths $[3, 3, 9, 3]$. (B) An individual ConvNeXt block, comprising a 7×7 stride-1 depthwise convolution, channels-last Layer Normalization, linear expansion from C to $4C$, GELU, linear projection from $4C$ to C , layer scaling, DropPath, and residual addition.

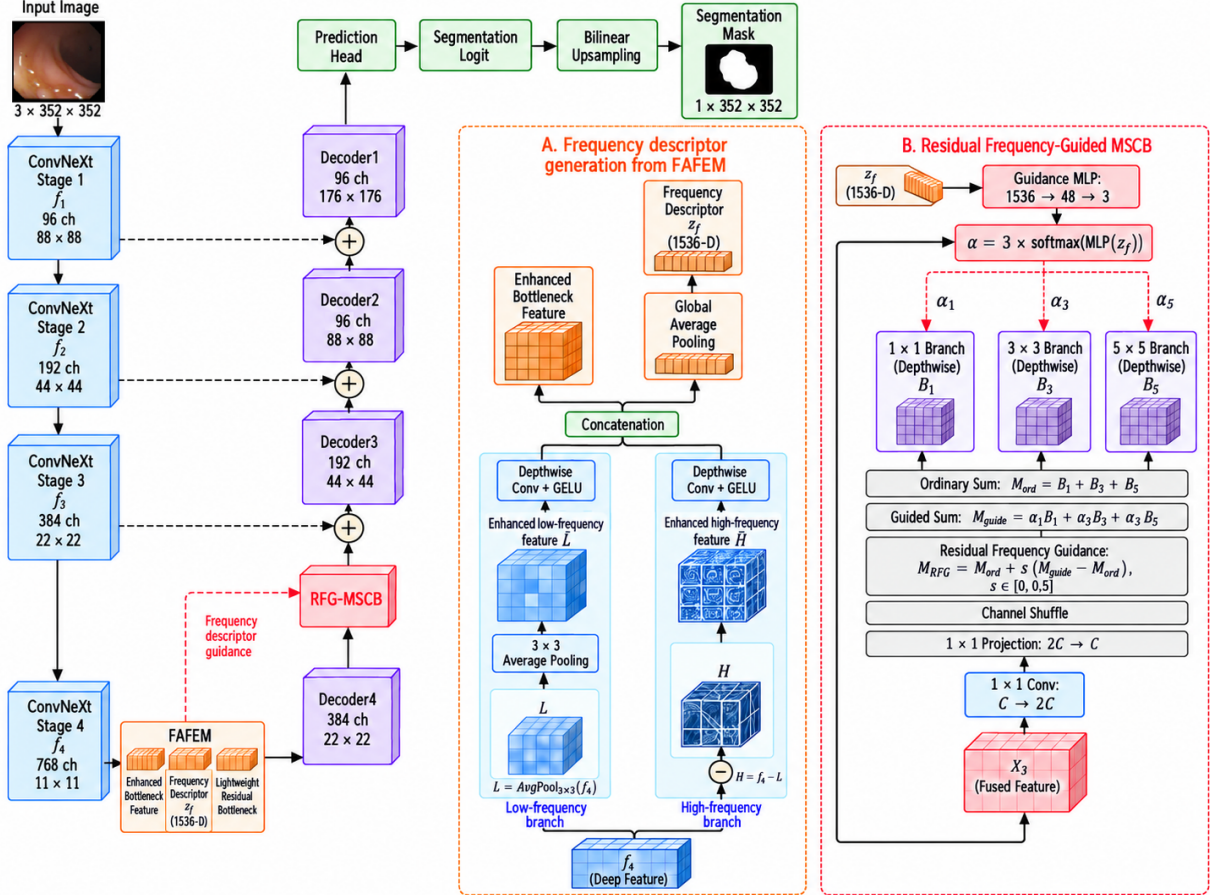


Figure 2: Overall architecture and detailed mechanism of the proposed segmentation framework. The left panel shows the ConvNeXt-Tiny encoder-decoder network with bottleneck FAFEM, Stage-3 RFG-MSCB, and the prediction head. The middle panel illustrates how FAFEM independently processes low- and high-frequency components of the deepest encoder feature, concatenates them, and generates the descriptor z_f . The right panel details RFG-MSCB, in which a lightweight guidance MLP uses z_f to adaptively weight the 1 × 1, 3 × 3, and 5 × 5 depthwise branches, and bounded residual coupling introduces the guided response as a deviation from ordinary MSCB aggregation.

$$X_L = \text{AvgPool}_{3 \times 3}(X), \quad (10)$$

$$X_H = X - X_L. \quad (11)$$

The low- and high-frequency components in Eqs. (10) and (11) are independently transformed using depth-wise convolutional branches,

$$\widehat{X}_L = \phi_L(X_L), \quad \widehat{X}_H = \phi_H(X_H), \quad (12)$$

where each $\phi(\cdot)$ contains a 3×3 depthwise convolution followed by GELU activation.

The two components are concatenated and globally pooled to obtain the frequency descriptor

$$\mathbf{z}_f = \text{GAP}([\widehat{X}_L, \widehat{X}_H]) \in \mathbb{R}^{2C}. \quad (13)$$

Since $C = 768$ at the bottleneck, $\mathbf{z}_f \in \mathbb{R}^{1536}$.

A lightweight channel-gating network applies global average pooling to $[\widehat{X}_L, \widehat{X}_H]$, followed by 1×1 transformations $1536 \rightarrow 48 \rightarrow 1536$, GELU after the first transformation, and a sigmoid after the second. Splitting the output channel-wise gives $w_L, w_H \in (0, 1)^{768}$; these weights are not normalized to sum to one. Their weighted combination is spatially refined and introduced through a residual connection:

$$\widetilde{F}_4 = F_4 + \gamma_f \mathcal{R}_f(w_L \odot \widehat{X}_L + w_H \odot \widehat{X}_H), \quad (14)$$

where $\mathcal{R}_f$ is a bias-free 3×3 depthwise convolution with stride 1 and padding 1, and γ_f is a learnable scalar initialized to zero. The descriptor $\mathbf{z}_f$ is pooled before the channel weights are applied.

3.4 Lightweight Residual Bottleneck

The frequency-enhanced representation $\widetilde{F}_4$ is subsequently processed by a lightweight residual bottleneck.

A 1×1 convolution first reduces the number of channels from 768 to 192:

$$X_r = \text{GELU}\left(\text{LN}\left(\text{Conv}_{1 \times 1}(\widetilde{F}_4)\right)\right). \quad (15)$$

A bias-free 3×3 depthwise-separable convolution with padding 1 performs spatial transformation in the reduced feature space, followed by channels-first Layer Normalization ($\epsilon = 10^{-6}$), GELU, and spatial dropout with probability 0.2. A bias-free 1×1 convolution then projects the representation back to 768 channels and is followed by channels-first Layer Normalization.

The bottleneck output is formulated as

$$B = \widetilde{F}_4 + \gamma_b \mathcal{H}(\widetilde{F}_4), \quad (16)$$

where $\mathcal{H}(\cdot)$ denotes the bottleneck transformation and γ_b is a learnable scalar initialized to zero.

This initialization makes the bottleneck behave initially as an identity mapping and allows its transformation to be progressively introduced during optimization.

3.5 Residual Frequency-Guided Multi-Scale Convolution

The principal architectural contribution of this work is the Residual Frequency-Guided Multi-Scale Convolution mechanism.

After the bottleneck feature is upsampled from $H/32$ to $H/16$, the decoder representation is reduced from 768 to 384 channels and concatenated with the Stage-3 encoder feature F_3 . A lightweight fusion block transforms the resulting 768-channel representation into $X_3 \in \mathbb{R}^{384 \times H/16 \times W/16}$, or $\mathbb{R}^{384 \times 22 \times 22}$ for a 352×352 input.

A conventional lightweight multi-scale convolution block first expands the feature dimension from C to $2C$:

$$X_e = \text{ReLU6}(\text{BN}(\text{Conv}_{1 \times 1}(X_3))). \quad (17)$$

Three parallel depthwise convolution branches with kernel sizes $k \in \{1, 3, 5\}$ then generate multi-scale responses,

$$B_k = \phi_k(X_e), \quad (18)$$

where each ϕ_k consists of a depthwise convolution, Batch Normalization, and ReLU6 activation.

In a conventional MSCB, these branches are combined uniformly:

$$M_{\text{ord}} = B_1 + B_3 + B_5. \quad (19)$$

The new mechanism conditions the relative contributions of the three receptive fields on the bottleneck frequency descriptor $\mathbf{z}_f$.

A lightweight guidance network maps the 1536-dimensional descriptor to three branch logits:

$$\mathbf{q} = W_{g2}(\text{GELU}(W_{g1}\mathbf{z}_f)), \quad (20)$$

where $W_{g1} : 1536 \rightarrow 48$ and $W_{g2} : 48 \rightarrow 3$.

The normalized branch weights are calculated as

$$[\alpha_1, \alpha_3, \alpha_5] = 3 \text{Softmax}(\mathbf{q}), \quad (21)$$

such that $\alpha_1 + \alpha_3 + \alpha_5 = 3$.

The frequency-guided response becomes

$$M_{\text{guide}} = \alpha_1 B_1 + \alpha_3 B_3 + \alpha_5 B_5. \quad (22)$$

We introduce the guidance as a bounded residual deviation from the conventional multi-scale response:

$$M_{\text{RFG}} = M_{\text{ord}} + s(M_{\text{guide}} - M_{\text{ord}}), \quad (23)$$

where $0 \leq s \leq 0.5$.

The implementation stores an unconstrained learnable scalar θ_s , initialized to 0.05, and uses the hard-clamped effective strength $s = \text{clamp}(\theta_s, 0, 0.5)$ in Eq. (23).

The final guidance-layer weights and biases are initialized so that the branch logits are equal. Thus, at initialization, $\alpha_1 = \alpha_3 = \alpha_5 = 1$ and $M_{\text{guide}} = M_{\text{ord}}$.

The module thus initially reproduces the conventional multi-scale path, while optimization is allowed to learn image-dependent deviations from uniform branch weighting.

The resulting feature is processed using channel shuffle with $\gcd(2C, C) = 384$ groups and a bias-free 1×1 projection from $2C$ back to C , followed by Batch Normalization. This sequence defines $\mathcal{P}$ below. A final residual connection preserves the input representation:

$$Y_3 = X_3 + \mathcal{P}(\text{Shuffle}(M_{\text{RFG}})). \quad (24)$$

This formulation connects the bottleneck frequency descriptor to multi-scale spatial refinement at $H/16$ resolution.

3.6 Progressive Decoder and Skip Fusion

After frequency-guided Stage-3 refinement, the decoder progressively restores spatial resolution using bilinear upsampling and depthwise-separable convolution.

The first refined decoder feature has 384 channels, $D_4 = Y_3$.

The subsequent decoder stages follow Eqs. (5)–(7).

For a 352×352 input, the decoder dimensions are $D_4 : 384 \times 22 \times 22$, $D_3 : 192 \times 44 \times 44$, $D_2 : 96 \times 88 \times 88$, and $D_1 : 96 \times 176 \times 176$.

At each skip level, the decoder feature is resized to the skip-feature shape when needed using bilinear interpolation with `align_corners=False`, concatenated with the corresponding encoder feature, and processed by a lightweight fusion block. The fusion block contains a bias-free 1×1 projection, channels-first Layer Normalization ($\epsilon = 10^{-6}$), GELU, a bias-free depthwise-separable 3×3 convolution with padding 1, a second channels-first Layer Normalization and GELU, and spatial dropout with probability 0.2. The primary twofold upsampling inside each decoder block uses bilinear interpolation with `align_corners=True`.

No additional attention mechanism, frequency module, MSCB, or boundary-specific refinement block is applied at these decoder stages.

The final decoder stage has no direct encoder skip connection because the ConvNeXt stem begins at one-quarter of the input resolution.

3.7 Prediction Head

The final decoder feature D_1 is processed using a lightweight refinement head. Two depthwise-separable 3×3 convolutions transform the feature dimensions as $96 \rightarrow 96 \rightarrow 48$.

Each transformation is followed by Layer Normalization and GELU activation. A final 1×1 convolution produces a single-channel segmentation logit map:

$$P_0 = \text{Conv}_{1 \times 1}(\mathcal{R}(D_1)), \quad (25)$$

$$P = \text{Up}_{H,W}(P_0), \quad (26)$$

$$\hat{Y} = \sigma(P). \quad (27)$$

The prediction in Eq. (26) is bilinearly resized from $H/2 \times W/2$ to the original input resolution with `align_corners=False`. The network directly returns logits; during inference, Eq. (27) gives the foreground probability map,

where $\sigma(\cdot)$ denotes the sigmoid function.

The selected architecture does not employ auxiliary prediction heads or deep supervision.

3.8 Model Complexity

The complete model has 30.057 M parameters and requires 27.818 GFLOPs for a 352×352 input. The FAFEM + MSCB configuration has 29.983 M parameters and 27.818 GFLOPs; the final RFG modification therefore adds about 0.074 M parameters, while GFLOPs are effectively unchanged at the displayed precision. Adaptive computation is localized to the Stage-3 encoder–decoder fusion, and the remaining decoder stages retain depthwise-separable convolution. This localization is empirically motivated by the MSCB-lite placement analysis in Table 4, rather than an exhaustive placement evaluation of Residual RFG-MSCB. Matched-hardware runtime efficiency for the four internal configurations is reported with the ablation results in Table 3.

Table 1: Computational complexity comparison. Parameter counts and computational cost are taken from original publications or official implementations where available; ‘–’ denotes an unavailable or unverified value.

Method	Year	Input	Params (M)	GFLOPs
MEGANet (Res2Net-50) [27]	2024	352^2	44.19	28.85
MSBP-Net [28]	2026	352^2	25.52	12.86
CIFFormer [15]	2025	352^2	32.44	13.14
PFPRNet [16]	2025	352^2	118.52	39.39
MLB-Net [10]	2026	352^2	87.70	–
MSPN [11]	2026	352^2	28.56	13.21
DVFIP-Net [12]	2026	352^2	45.62	19.37
Ours	–	352^2	30.057	27.818 [†]

[†]Our value uses 13.91 GMACs and the explicit convention $\text{FLOPs} = 2 \times \text{MACs}$. Competitor values are preserved in source terminology and are not convention-normalized.

4 Experiments

4.1 Datasets and Data Splits

To evaluate the proposed framework under a fixed training and evaluation design, we used Kvasir-SEG [31], CVC-ClinicDB [32], CVC-300 [33], CVC-ColonDB [34], and ETIS-LaribPolypDB [35]. Following a PraNet-style data partition, 100 of the 1000 Kvasir-SEG images and 62 of the 612 CVC-ClinicDB images were reserved as final seen test sets. The remaining 900 Kvasir-SEG and 550 CVC-ClinicDB images formed a combined 1450-image development pool. This pool was split deterministically into 1305 training images and 145 internal validation images using shuffled sampling with `random_state=42`; dataset membership was identical across all training seeds.

The final Kvasir-SEG and CVC-ClinicDB test images were excluded from training, validation, checkpoint selection, early stopping, threshold tuning, hyperparameter tuning, and architecture selection. CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB were used exclusively as external test sets and did not influence model selection.

Table 2: Dataset partition used in the experiments.

Dataset	Total	Development pool	Test
Kvasir-SEG	1000	900	100
CVC-ClinicDB	612	550	62
CVC-300	60	–	60
CVC-ColonDB	380	–	380
ETIS-LaribPolypDB	196	–	196

Images were converted to RGB, resized to 352×352 with bilinear interpolation, and scaled to $[0, 1]$. Masks

were resized with nearest-neighbor interpolation and binarized. The encoder applied ImageNet normalization with mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225).

4.2 Implementation Details

The final protocol uses seeds 42, 6543, and 7777. All final and ablation results are reported as the mean $\pm$ sample standard deviation (ddof=1) across these seeds unless otherwise noted. The runs were implemented in PyTorch 2.5.1+cu124 on Windows and executed on an NVIDIA GeForce RTX 4090, with a maximum of 200 epochs and batch size 24. AdamW with betas (0.9, 0.999) used separate parameter groups. The maximum learning rates were 3.0×10^{-4} for the decoder and 4.5×10^{-4} for the final refinement head. Encoder Stages 4, 3, 2, and 1 plus the stem used 3.0×10^{-5} , 2.4×10^{-5} , 1.92×10^{-5} , and 1.536×10^{-5} , respectively, corresponding to layer-wise decay 0.8. Weight decay was 1.0×10^{-4} for the decoder, 5.0×10^{-2} for the encoder, and 1.0×10^{-2} for the refinement head.

Learning rates were warmed up linearly for five epochs from 0.1 to 1.0 of each group maximum and then cosine-annealed over the remaining epochs to 1.0×10^{-6} . Training used AMP FP16, global gradient clipping at norm 1.0, and an exponential moving average (EMA) with decay 0.995. All parameters were trainable from the first epoch; no staged freezing or decoder-only warmup was used. The encoder was initialized from `convnext_tiny_22k_1k_384.pth`, logged as ImageNet pretrained weights. Deep supervision and frequency augmentation were disabled, and sampling was uniform. Early stopping was configured with patience 30 from the start of training. Seeds 42 and 6543 completed the maximum 200 epochs, corresponding to zero-based epochs 0–199; seed 7777 stopped after epoch 167 (168 completed epochs). Model complexity and its measurement convention are reported in Table 1.

Training augmentation applied horizontal and vertical flips independently with probability 0.5, rotation in $[-15^\circ, 15^\circ]$ with probability 0.5, and a random crop ratio in $[0.7, 1.0]$ followed by resizing with probability 0.5. Joint brightness and contrast factors in $[0.85, 1.15]$ and saturation in $[0.9, 1.1]$ were applied with probability 0.5. A 3×3 Gaussian blur and additive Gaussian noise with standard deviation 0.015 were applied with probabilities 0.15 and 0.25, respectively.

The code seeded Python, NumPy, PyTorch CPU, and all CUDA random-number generators. Deterministic CUDA algorithms and cuDNN deterministic backend flags were not enabled.

4.3 Loss Function

The final objective was

$$\mathcal{L} = 0.55\mathcal{L}_{\text{Dice}} + 0.25\mathcal{L}_{\text{BCE}} + 0.20\mathcal{L}_{\text{boundary}}. \quad (28)$$

The objective in Eq. (28) uses a soft Dice term with a smoothing constant of 1.0. The BCE-with-logits term used targets smoothed as $y_{\text{BCE}} = 0.98y + 0.01$, corresponding to label smoothing 0.02. For the boundary-aware term, fixed Sobel filters were applied to each binary mask to obtain $g = \sqrt{g_x^2 + g_y^2}$. The magnitude was normalized per image as $a = g / \max(\max(g), 10^{-6})$, and the pixel weights were $w = 1 + \kappa a$ with $\kappa = 5$; consequently, the weights range from 1 to 6. For probability $p = \sigma(P)$ and target y , the term was

$$\mathcal{L}_{\text{boundary}} = -\frac{\sum_i w_i [y_i \log(p_i + \epsilon) + (1 - y_i) \log(1 - p_i + \epsilon)]}{\sum_i w_i + \epsilon} + 1 - \frac{\sum_i w_i y_i p_i + \epsilon}{\sum_i w_i (y_i + p_i - y_i p_i) + \epsilon}, \quad (29)$$

with $\epsilon = 10^{-6}$ and batch averaging for the weighted IoU component. The focal-Tversky term was assigned zero weight and therefore did not contribute to the final objective.

4.4 Evaluation Metrics

We report mean Dice (mDice), mean intersection over union (mIoU), S-measure S_α [36], weighted F-measure F_β^w [37], mean and maximum E-measure (mE_ϕ and $\max E_\phi$) [38], and mean absolute error (MAE) [39]. For each image, the thresholded prediction P and binary ground truth G were flattened and evaluated as

$$\text{Dice}(P, G) = \frac{2|P \cap G| + 10^{-6}}{|P| + |G| + 10^{-6}}, \quad \text{IoU}(P, G) = \frac{|P \cap G| + 10^{-6}}{|P| + |G| - |P \cap G| + 10^{-6}}. \quad (30)$$

The reported mDice and mIoU are arithmetic means of the per-image scores in Eq. (30), rather than global pixel aggregates. S-measure, weighted F-measure, the E-measure curve, and MAE were computed per image with `py_sod_metrics` from 8-bit probability maps and binary ground truth. The mean and maximum of the resulting E-measure curve were reported.

4.5 Model Selection and Evaluation Protocol

At each epoch, EMA weights were evaluated only on the 145-image internal validation subset. Probabilities were obtained with one sigmoid operation and binarized using the strict rule $p > 0.45$. A checkpoint was saved only when internal-validation IoU was strictly greater than the previous best. Best checkpoints were selected independently for each seed according to internal-validation IoU.

After model selection, each best checkpoint was loaded strictly and evaluated on the 100-image Kvasir-SEG and 62-image CVC-ClinicDB seen test sets, followed by the full CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB external test sets. Test-time augmentation was disabled. Neither the seen nor external test results influenced checkpoint selection or other model-development decisions.

5 Experimental Study

The experimental study first examines the contribution of the architectural components under controlled variants and then compares the resulting model with published methods. Presenting the ablation before the comparative evaluation establishes the reference configuration and clarifies which part of the final design is being assessed in the comparative tables.

5.1 Ablation Study

The ablation design compares four configurations in sequence: Baseline, FAFEM, FAFEM + MSCB, and Residual RFG-MSCB (Ours). This sequence first isolates bottleneck frequency-aware enhancement, then adds Stage-3 multi-scale refinement, and finally assesses the residual frequency-guided modification.

5.1.1 Ablation Results

The Baseline serves as the reference configuration without FAFEM and MSCB. Adding FAFEM introduces bottleneck frequency-aware enhancement, while FAFEM + MSCB further adds Stage-3 multi-scale convolutional refinement. Residual RFG-MSCB then introduces cross-level frequency-guided coupling, allowing the bottleneck descriptor to modulate the multi-scale branches at the Stage-3 decoder fusion. Table 3 reports mean $\pm$ sample standard deviation across three seed-level macro averages. Each seed-level value is the unweighted arithmetic mean across Kvasir-SEG, CVC-ClinicDB, CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB.

5.1.2 Ablation Interpretation

Table 3 presents the macro-averaged ablation results across the five polyp segmentation datasets. Relative to the Baseline, FAFEM increases mDice and mIoU by 3.7 and 3.5 percentage points, respectively. The corresponding changes are +3.0 points in F_β^w , +1.8 in S_α , +2.7 in mE_ϕ , and +2.9 in $\max E_\phi$, while MAE decreases by 0.1 point from 0.019 to 0.018.

Adding MSCB to FAFEM raises mDice, mIoU, and F_β^w by 0.3, 0.5, and 0.4 percentage points, respectively. The increases in S_α , mE_ϕ , and $\max E_\phi$ are 0.2, 0.3, and 0.1 points, while the displayed MAE remains 0.018.

Relative to FAFEM + MSCB, Residual RFG-MSCB adds 0.5 points in mDice, 0.8 in mIoU, 0.4 in F_β^w , 0.4 in S_α , 0.4 in mE_ϕ , and 0.3 in $\max E_\phi$. It consequently has the highest mean for each of these six metrics: 0.872, 0.810, 0.849, 0.914, 0.941, and 0.949, respectively. Its displayed MAE remains 0.018, tied with FAFEM and FAFEM + MSCB.

Table 3: Ablation and matched-hardware efficiency comparison across four model configurations. Segmentation metrics are five-dataset macro averages reported as mean $\pm$ sample standard deviation across three seeds. Runtime measurements used an NVIDIA GeForce RTX 4090, FP16 inference, batch size 1, and 352×352 inputs.

Model	mDice	mIoU	F_{β}^w	S_{α}	mE_{ϕ}	$\max E_{\phi}$	MAE	Params (M)	GFLOPs	Latency (ms)	FPS	Peak VRAM (MiB)
Baseline	0.827 ± 0.007	0.762 ± 0.008	0.811 ± 0.006	0.890 ± 0.004	0.907 ± 0.005	0.916 ± 0.005	0.019 ± 0.000	29.190	27.202	10.573 ± 0.196	94.585	248.859
FAFEM	0.864 ± 0.002	0.797 ± 0.001	0.841 ± 0.003	0.908 ± 0.001	0.934 ± 0.002	0.945 ± 0.002	0.018 ± 0.000	29.360	27.208	11.014 ± 0.137	90.794	249.832
FAFEM + MSCB	0.867 ± 0.002	0.802 ± 0.003	0.845 ± 0.003	0.910 ± 0.002	0.937 ± 0.001	0.946 ± 0.001	0.018 ± 0.000	29.983	27.818	11.630 ± 0.137	85.986	252.876
Residual RFG-MSCB (Ours)	0.872 ± 0.004	0.810 ± 0.004	0.849 ± 0.003	0.914 ± 0.003	0.941 ± 0.001	0.949 ± 0.001	0.018 ± 0.000	30.057	27.818	11.806 ± 0.023	84.701	253.306

Runtime protocol: PyTorch 2.5.1+cu124 and CUDA 12.4 on an NVIDIA GeForce RTX 4090; FP16 autocast, batch size 1, $1 \times 3 \times 352 \times 352$ input, forward pass only, and no TTA, I/O, or post-processing. Each of three repetitions used 100 warm-ups and 500 synchronized CUDA-event measurements. Latency variation is the sample standard deviation of the three repetition means.

The progression increases parameters from 29.190 M for the Baseline to 29.360 M with FAFEM, 29.983 M with MSCB, and 30.057 M with Residual RFG-MSCB. GFLOPs increase from 27.202 to 27.208 and 27.818 after adding MSCB, then remain 27.818 for the final configuration at the displayed precision. Thus the complete configuration adds 0.867 M parameters and 0.616 GFLOPs over the Baseline. Relative to FAFEM + MSCB, the residual frequency-guided extension adds about 0.074 M parameters while leaving GFLOPs unchanged at the displayed precision. Under matched FP16 inference, latency increases from 11.630 to 11.806 ms (1.52%), FPS decreases from 85.986 to 84.701 (1.50%), and peak allocated VRAM increases from 252.876 to 253.306 MiB (0.17%). This represents modest measured runtime overhead, with throughput remaining 84.70 FPS. Together with the metric changes, this four-configuration progression supports positive descriptive contributions from each added component at the reported aggregate level. No statistical significance is claimed.

5.2 MSCB-lite Placement Analysis

Since Table 3 identifies multi-scale refinement as an effective component, we further examined whether extending MSCB-lite beyond the Stage-3 fusion provides additional benefit. The matched comparison evaluates refinement at Stage 3, Stages 3 + 2, and Stages 3 + 2 + 1 using the same five-dataset macro-averaging procedure. The overlap metrics are summarized across the evaluated seeds.

Table 4: Placement analysis of MSCB-lite using the same five-dataset macro-averaging procedure. Values are reported as mean $\pm$ sample standard deviation across the evaluated seeds.

Placement	mDice	mIoU	Params (M)	GFLOPs
Stage 3	0.8685 ± 0.006	0.8027 ± 0.005	29.983	27.818
Stages 3 + 2	0.8640 ± 0.007	0.7985 ± 0.003	30.148	28.468
Stages 3 + 2 + 1	0.8660 ± 0.005	0.8014 ± 0.002	30.193	29.197

Stage 3 alone has the highest mean mDice (0.8685) and mean mIoU (0.8027), as well as the lowest parameter count (29.983 M) and GFLOPs (27.818). Extending MSCB-lite to additional stages does not improve the mean overlap metrics, while parameters increase to 30.148 M and 30.193 M and GFLOPs increase to 28.468 and 29.197. Stage 3 therefore provides the most favorable observed accuracy–complexity trade-off among the tested MSCB-lite placements. These descriptive comparisons do not establish statistical significance. This placement behavior provides empirical motivation for retaining multi-scale refinement at Stage 3 when introducing the residual frequency-guided formulation, while avoiding the additional cost of distributing

MSCB across shallower decoder levels. Residual RFG-MSCB itself was not exhaustively tested at alternative decoder-stage placements.

The qualitative examples in Fig. 3 complement the aggregate ablation results in Table 3 by illustrating how the architectural changes affect predicted mask geometry in the displayed samples.

5.3 Qualitative Ablation Analysis

Figure 3 shows the Baseline, FAFEM, FAFEM + MSCB, and Residual RFG-MSCB predictions in the same order as the quantitative ablation. The final column provides an additional spatial visualization associated with the final prediction. Because the original heatmap-generation metadata are not available in the current experiment artifacts, we use the visualization only as a qualitative spatial reference and do not assign it a specific interpretability mechanism. The added components alter mask coverage and boundary position across the displayed examples. In the CVC-ColonDB case with an elongated annotation, the disconnected Baseline regions are replaced by more coherent predictions from FAFEM, FAFEM + MSCB, and Residual RFG-MSCB.

Residual RFG-MSCB yields a more coherent or better-localized mask in selected cases, including the displayed CVC-ColonDB case, while its effect varies among the selected images. In the displayed Kvasir-SEG, CVC-ClinicDB, CVC-300, and small-lesion ETIS examples, the four configurations differ mainly in boundary position and extent, and no configuration reproduces every ground-truth contour exactly. These examples therefore provide a visual complement to the aggregate trends in Table 3, rather than evidence of uniform improvement for every image.

5.4 Comparative Study

After the component-level analysis, the comparative study evaluates the complete model on the Kvasir-SEG and CVC-ClinicDB seen test sets and on three external test sets, and places its performance alongside published baselines. The quantitative results are presented below.

5.4.1 Quantitative Analysis

On Kvasir-SEG, the proposed model achieved the highest displayed mDice and mIoU, reaching 0.932 ± 0.0044 and 0.883 ± 0.004 , respectively. PFPRNet was second-highest for both metrics, with mDice of 0.930 and mIoU of 0.881. On CVC-ClinicDB, the proposed model achieved 0.948 ± 0.003 mDice, which was second-highest to PFPRNet’s 0.949, and the highest displayed mIoU at 0.904 ± 0.004 . It also tied for the highest displayed S_α on CVC-ClinicDB at 0.958 ± 0.001 and was second-highest in mE_ϕ and $\max E_\phi$ at 0.979 ± 0.001 and 0.988 ± 0.002 , respectively.

The main comparison is restricted to methods using comparable PraNet-style data protocols with closely matched Kvasir-SEG and CVC-ClinicDB partitioning. Methods using materially different data partitions remain discussed in Related Work but are omitted from the main performance tables. Differences in optimization, augmentation, backbone initialization, model-selection criteria, and implementation details remain. Across the comparison tables, the best and second-best displayed values are shown in bold and underlined, respectively.

On the external datasets, mDice/mIoU were $0.916 \pm 0.002/0.863 \pm 0.002$ on CVC-300, $0.824 \pm 0.009/0.747 \pm 0.008$ on CVC-ColonDB, and $0.816 \pm 0.012/0.763 \pm 0.013$ on ETIS-LaribPolypDB. Across these three external datasets, the model’s mDice ranged from 0.816 to 0.916 and its mIoU ranged from 0.747 to 0.863; the largest between-seed variation in both metrics occurred on ETIS. On CVC-300, the model achieved the highest displayed mDice, mIoU, S_α , and $\max E_\phi$ and tied for the lowest displayed MAE. Its mIoU, mE_ϕ , and $\max E_\phi$ were second-highest on ETIS, while its mIoU and $\max E_\phi$ were second-highest and its MAE was second-lowest on CVC-ColonDB.

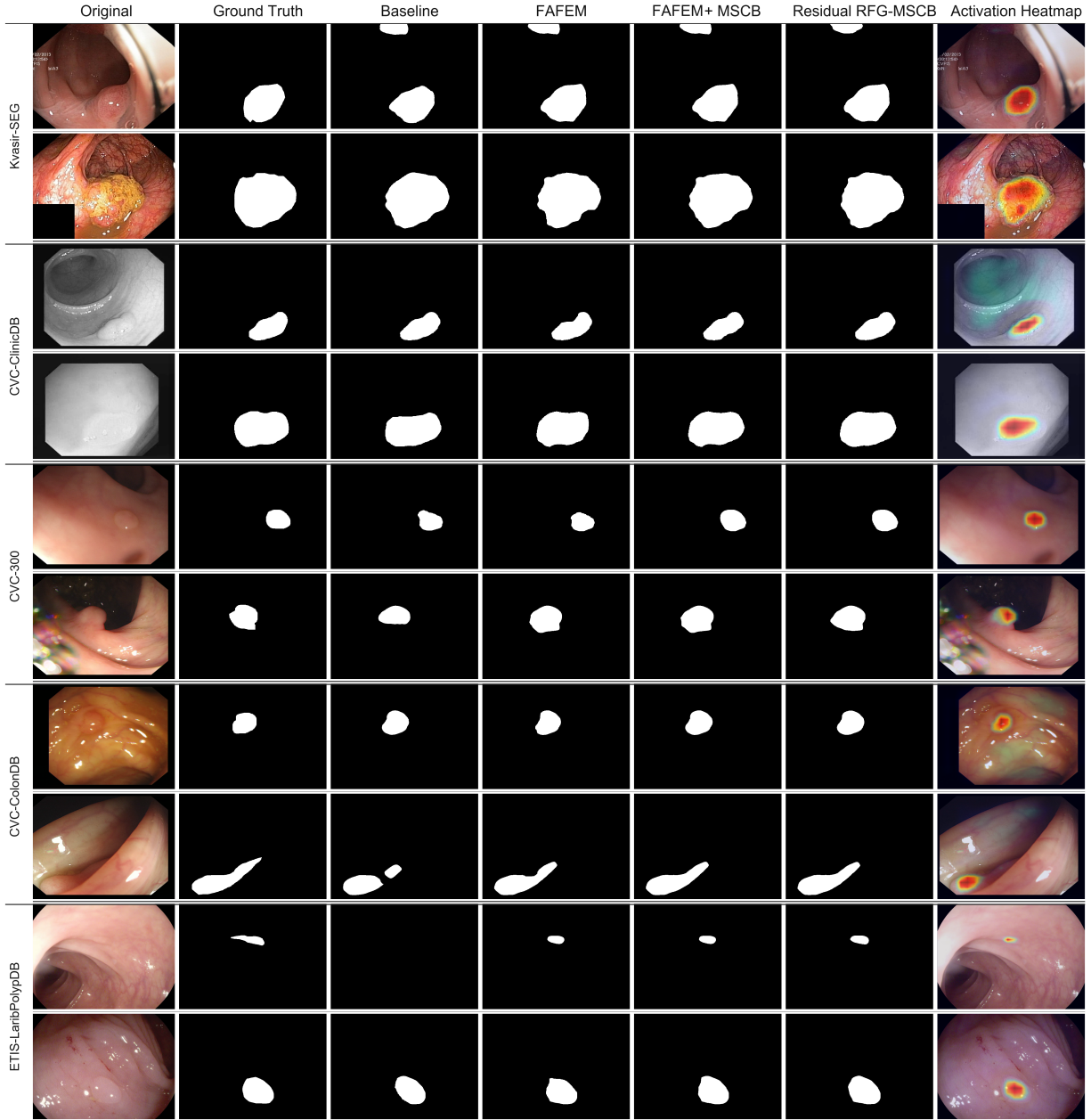


Figure 3: Qualitative comparison of the ablation configurations using two displayed samples per dataset. Dataset groups correspond to Kvasir-SEG, CVC-ClinicDB, CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB. Columns show, from left to right, the original image, ground-truth mask, Baseline prediction, FAFEM prediction, FAFEM + MSCB prediction, Residual RFG-MSCB prediction, and the corresponding activation heatmap.

Table 5: Quantitative comparison on Kvasir-SEG.

Methods	Year	mDice	mIoU	F_{β}^w	S_{α}	mE_{ϕ}	$\max E_{\phi}$	MAE
MEGANet [27]	2024	0.913	0.863	0.907	0.918	0.956	0.959	0.025
MSBP-Net [28]	2026	0.919	0.868	0.916	0.926	0.961	—	0.023
CIFFormer [15]	2025	0.926	0.876	0.929	0.939	0.967	—	0.019
PFPRNet [16]	2025	<u>0.930</u>	<u>0.881</u>	0.920	0.935	0.971	—	0.019
MLB-Net [10]	2026	0.926	0.878	<u>0.921</u>	<u>0.938</u>	<u>0.969</u>	0.975	0.021
MSPN [11]	2026	0.927	0.879	—	—	—	—	0.021
DVFIP-Net [12]	2026	0.922	0.874	—	—	—	—	—
Ours	—	0.932	0.883	0.914	0.936	0.962	0.974	0.020
		± 0.0044	± 0.004	± 0.006	± 0.003	± 0.003	± 0.002	± 0.001

Table 6: Quantitative comparison on CVC-ClinicDB.

Methods	Year	mDice	mIoU	F_{β}^w	S_{α}	mE_{ϕ}	$\max E_{\phi}$	MAE
MEGANet [27]	2024	0.938	0.894	0.940	0.950	—	0.987	<u>0.007</u>
MSBP-Net [28]	2026	0.940	0.892	0.927	0.939	0.976	—	<u>0.007</u>
CIFFormer [15]	2025	0.944	0.897	0.941	0.934	0.972	—	<u>0.007</u>
PFPRNet [16]	2025	0.949	<u>0.903</u>	<u>0.947</u>	<u>0.957</u>	0.989	—	0.006
MLB-Net [10]	2026	0.945	0.902	0.948	0.958	0.989	0.992	0.006
MSPN [11]	2026	0.943	0.902	—	—	—	—	0.006
DVFIP-Net [12]	2026	0.942	0.896	—	—	—	—	—
Ours	—	0.948	0.904	0.939	0.958	0.979	0.988	0.007
		± 0.003	± 0.004	± 0.001	± 0.001	± 0.001	± 0.002	± 0.001

Table 7: Quantitative comparison on ETIS-LaribPolypDB.

Methods	Year	mDice	mIoU	F_{β}^w	S_{α}	mE_{ϕ}	$\max E_{\phi}$	MAE
MEGANet [27]	2024	0.789	0.709	0.753	0.866	0.855	0.915	0.015
MSBP-Net [28]	2026	0.795	0.718	0.770	0.873	0.899	—	0.019
CIFFormer [15]	2025	0.810	0.746	0.858	0.889	0.918	—	<u>0.013</u>
PFPRNet [16]	2025	0.828	0.764	<u>0.839</u>	0.908	0.935	—	0.012
MLB-Net [10]	2026	<u>0.821</u>	0.744	0.780	<u>0.899</u>	0.914	0.935	<u>0.013</u>
MSPN [11]	2026	0.798	0.717	—	—	—	—	0.017
DVFIP-Net [12]	2026	0.811	0.730	—	—	—	—	—
Ours	—	0.816	0.763	0.783	0.882	0.921	0.927	0.014
		± 0.012	± 0.013	± 0.004	± 0.001	± 0.006	± 0.010	± 0.001

Table 8: Quantitative comparison on CVC-ColonDB.

Methods	Year	mDice	mIoU	F_{β}^w	S_{α}	mE_{ϕ}	$\max E_{\phi}$	MAE
MEGANet [27]	2024	0.793	0.714	0.779	0.854	0.892	0.895	0.034
MSBP-Net [28]	2026	0.810	0.731	0.810	0.861	0.912	—	0.029
CIFFormer [15]	2025	0.823	0.741	0.851	0.874	<u>0.918</u>	—	0.026
PFPRNet [16]	2025	0.819	0.741	<u>0.843</u>	<u>0.882</u>	0.915	—	0.028

Continued on next page

Table 8: Quantitative comparison on CVC-ColonDB (continued).

Methods	Year	mDice	mIoU	F_{β}^w	S_{α}	mE_{ϕ}	$\max E_{\phi}$	MAE
MLB-Net [10]	2026	0.832	0.757	0.825	0.886	0.924	0.931	0.023
MSPN [11]	2026	0.811	0.734	—	—	—	—	0.029
DVFIP-Net [12]	2026	<u>0.826</u>	0.746	—	—	—	—	—
Ours	—	0.824 ± 0.009	0.747 ± 0.008	0.832 ± 0.002	0.862 ± 0.004	0.911 ± 0.003	0.919 ± 0.002	0.025 ± 0.001

Table 9: Quantitative comparison on CVC-300 (EndoScene).

Methods	Year	mDice	mIoU	F_{β}^w	S_{α}	mE_{ϕ}	$\max E_{\phi}$	MAE
MEGANet [27]	2024	0.899	0.834	0.882	0.935	0.966	0.969	0.007
MSBP-Net [28]	2026	0.903	<u>0.859</u>	0.927	0.942	0.967	—	0.019
CIFFormer [15]	2025	0.883	0.822	0.912	0.939	0.961	—	0.100
PFPRNet [16]	2025	0.891	0.833	<u>0.917</u>	0.928	0.950	—	0.013
MLB-Net [10]	2026	<u>0.915</u>	0.852	<u>0.909</u>	<u>0.974</u>	0.974	<u>0.978</u>	0.005
MSPN [11]	2026	0.907	0.840	—	—	—	—	<u>0.006</u>
DVFIP-Net [12]	2026	0.900	0.833	—	—	—	—	—
Ours	—	0.916 ± 0.002	0.863 ± 0.002	0.847 ± 0.005	0.975 ± 0.004	0.969 ± 0.004	0.979 ± 0.005	0.005 ± 0.001

6 Discussion

6.1 Component Contributions

The evaluation protocol separates seen-domain testing on Kvasir-SEG and CVC-ClinicDB from external testing on CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB. All five test sets are isolated from checkpoint selection, which uses only the internal validation subset of the combined development pool.

The controlled ablation evaluates Baseline, FAFEM, FAFEM + MSCB, and Residual RFG-MSCB using unweighted five-dataset macro averages and three training seeds. FAFEM raises mean mDice from 0.827 to 0.864, a 3.7-percentage-point increase, and adding MSCB raises it by a further 0.3 points to 0.867. Residual RFG-MSCB then adds 0.5 points to reach 0.872 mDice and has the highest means for mDice, mIoU, F_{β}^w , S_{α} , mE_{ϕ} , and $\max E_{\phi}$. The displayed MAE decreases from 0.019 to 0.018 with FAFEM and remains tied at 0.018 for the two subsequent configurations. These descriptive aggregate changes support the combined residual frequency-guided formulation at the Stage-3 fusion without establishing improvement for every image or statistical significance.

The bounded residual design preserves the conventional MSCB response while allowing sample-dependent branch modulation. Because residualization and frequency conditioning were not evaluated independently, their individual contributions remain unresolved. This adaptive refinement reuses the bottleneck descriptor at one decoder stage without discarding the established multi-scale response or modifying the full decoder.

6.2 Cross-Dataset Generalization

CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB are reserved for external generalization evaluation and do not influence model development. Differences in acquisition, annotation, case mix, and polyp morphology make these datasets a distinct test of transfer beyond the Kvasir-SEG/CVC-ClinicDB development domain.

On the external datasets, the model obtained mDice/mIoU of 0.916/0.863 on CVC-300, 0.824/0.747 on CVC-ColonDB, and 0.816/0.763 on ETIS-LaribPolypDB. Performance varied across these datasets, with the strongest transfer observed on CVC-300. On CVC-ColonDB and ETIS, the model’s mIoU was second-highest among the displayed methods, whereas its mDice was below the two highest displayed values. Differences in acquisition, annotation, and lesion appearance are plausible contributors to the lower CVC-ColonDB and ETIS results, but were not isolated experimentally.

Although the main comparison is restricted to methods with closely matched dataset partitioning, differences in optimization, augmentation, backbone initialization, model-selection criteria, and implementation details remain.

6.3 Practical Trade-offs and Placement

As summarized in Table 1, the final architecture has 30.057 M parameters and requires 27.818 GFLOPs at 352×352 . The matched-hardware results in Table 3 show modest runtime and memory overhead for the residual frequency-guided extension relative to FAFEM + MSCB. The Stage-3 localization is also consistent with the MSCB-lite placement analysis, where Stage 3 gave the best observed accuracy–complexity trade-off and extending refinement to additional skip levels increased computational cost without improving the five-dataset macro-average overlap metrics. This provides empirical motivation for Stage-3 localization but does not establish the optimal placement of Residual RFG-MSCB, which was not exhaustively placement-tested. Deployment-oriented latency optimization remains future work.

6.4 Limitations

- **No statistical significance testing:** All reported differences are descriptive; no hypothesis testing was performed.
- **Seed coverage:** The Table 3 comparisons use three training seeds, which limits characterization of run-to-run variability.
- **Component isolation:** Residualization and frequency conditioning were not independently disentangled.
- **Placement evaluation:** MSCB-lite placement was evaluated across the reported seeds, whereas Residual RFG-MSCB placement was not exhaustively tested across decoder stages or multiple seeds.
- **Runtime evaluation:** Runtime was measured on one GPU under one FP16, batch-size-1 inference setting; broader deployment evaluation across hardware platforms remains future work.
- **External generalization:** Performance was lower on CVC-ColonDB and ETIS-LaribPolypDB than on CVC-300, motivating broader evaluation across acquisition domains.

6.5 Future Work

- Systematic multi-seed placement evaluation of Residual RFG-MSCB with controlled batch size and learning rate.
- Disentanglement of residualization, frequency conditioning, and initialization through factorial experiments.
- Domain adaptation or targeted augmentation for CVC-ColonDB and ETIS-LaribPolypDB.
- Evaluation of between-configuration differences using larger seed ensembles or bootstrap confidence intervals.

7 Conclusion

We presented a ConvNeXt-Tiny U-shaped segmentation framework that couples a locally implemented bottleneck FAFEM descriptor to MSCB through RFG-MSCB at the Stage-3 encoder–decoder fusion. The design extends frequency enhancement into decoder refinement while preserving the ordinary multi-scale pathway as a reference.

The final model achieved mDice/mIoU of 0.932/0.883 on Kvasir-SEG and transferred most strongly to CVC-300 among the three external datasets, with lower performance on CVC-ColonDB and ETIS-LaribPolypDB. In the controlled ablation, FAFEM + MSCB reached 0.867 mDice before Residual RFG-MSCB increased the mean to 0.872 and achieved the highest means across all six positive-valued aggregate metrics. The displayed MAE remained 0.018 after the FAFEM stage. The final modification adds about 0.074 M parameters over FAFEM + MSCB with effectively unchanged GFLOPs at the displayed precision and modest measured runtime overhead, without changing the remaining depthwise-separable decoder stages.

Future work should evaluate Residual RFG-MSCB placement across multiple seeds, disentangle residualization from frequency conditioning, and examine adaptation across acquisition domains.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets used in this study are publicly available from the sources cited in the manuscript. The experiment implementation is available at <https://github.com/shosseinj/convnext-unet>.

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